

Project Report

On

Marketing Campaign Performance Analysis Dashboard

The Marketing Campaign Analysis project is designed to analyse customer-related marketing data and convert it into meaningful insights using Power BI.

"From Data to Decisions"

"Measure. Analyse. Optimize."

"Turning Campaign Data into Actionable Insights"

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Power BI | DAX | Excel | Power Query

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1 Project Title

Marketing Campaign Performance Analysis Dashboard

2 Problem Statement

Marketing campaigns generate large amounts of customer-related data, including demographics, spending behaviour, purchase channels, and campaign responses. However, many organizations find it difficult to analyse this data effectively using traditional manual methods. Spreadsheet-based analysis is time-consuming, prone to errors, and unsuitable for large and complex datasets. The lack of interactive and centralized reporting systems further limits the ability to explore data dynamically and evaluate campaign performance. In addition, unclean and unstructured data reduces the accuracy of analysis and makes it difficult to identify meaningful insights.

Therefore, there is a need for an efficient data analytics solution that can transform cleaned marketing data into an interactive and visually understandable dashboard. This project uses Power BI to analyse marketing campaign data and support data-driven decision-making.

3 Project Description

The Marketing Campaign Analysis project is designed to analyse customer-related marketing data and convert it into meaningful insights using Power BI. The project focuses on understanding customer behaviour, spending patterns, purchase channels, and responses to various marketing campaigns. By utilizing cleaned and structured data, the project aims to support data-driven decision-making and improve the effectiveness of marketing strategies.

In modern organizations, marketing departments collect large volumes of customer data from multiple sources. However, without proper analytics and visualization tools, this data remains underutilized. The primary objective of this project is to bridge the gap between raw marketing data and actionable business insights by developing an interactive dashboard that presents key metrics in a simple and visually appealing manner.

The project uses cleaned marketing campaign data as input. The data includes customer demographic information such as education qualification, marital status, income, country, and family structure, along with spending details across different product categories. It also contains information related to marketing campaign acceptance and purchase behaviour through different channels such as web, store, and catalog. This data is processed and analysed using Power BI to generate insights that can help organizations evaluate customer segments and campaign performance.

To ensure efficient analysis, the project follows a structured data modelling approach. Instead of using a single flat table, the cleaned data is logically divided into multiple tables, including Customer, Spending, Campaign, and Purchase Channel tables. These tables are connected using appropriate relationships based on a unique customer identifier. This approach improves data organization, enhances performance, and allows accurate cross-filtering between different aspects of the analysis.

The project workflow begins with loading the cleaned dataset into Power BI using Power Query. Power Query is used to transform and organize the data, remove unnecessary columns, and create calculated fields such as total spending. After data preparation, relationships are

established between the tables to form a structured data model. This data model serves as the foundation for all visualizations and analyses performed in the dashboard.

The dashboard developed as part of this project presents key performance indicators (KPIs) such as total customers, total spending, average income, and campaign response rate. These KPIs provide a quick overview of the overall marketing performance. In addition, various charts and graphs are used to analyse customer demographics, spending by product category, campaign responses, and purchase channel preferences. The use of visual analytics makes it easier for users to identify trends, patterns, and anomalies in the data.

Interactivity is an important aspect of the project. Filters applied across all pages allow users to analyse data based on parameters such as country, education qualification, marital status, and income range. This enables users to focus on specific customer segments and gain deeper insights into their behaviour and preferences. Such interactive analysis supports more informed and targeted marketing decisions.

The project does not focus on predictive modelling or real-time data processing. Instead, it emphasizes descriptive and diagnostic analytics, which are well-suited for understanding historical marketing performance. By analysing past campaign data, organizations can identify which customer groups are more responsive and which product categories generate higher spending.

Overall, this project demonstrates the practical application of business intelligence concepts using Power BI. It highlights how cleaned marketing data can be transformed into an interactive dashboard that supports strategic decision-making. The project serves as an effective example of how data visualization and analytics tools can be used to improve marketing performance, reduce manual analysis effort, and enhance business understanding.

3.1 Scope of the Work

The scope of this project defines the boundaries and extent of the work carried out for the Marketing Campaign Analysis using Power BI. The project primarily focuses on analysing cleaned marketing data to extract meaningful insights related to customer behaviour, spending patterns, and campaign performance. The scope is limited to descriptive and analytical reporting and does not include predictive modelling or real-time data processing.

The project covers the analysis of customer demographic information such as education level, marital status, income range, and country. These attributes are used to segment customers and understand how different demographic groups behave in terms of spending and campaign acceptance. The scope also includes the analysis of customer purchasing behaviour across various product categories, including wines, fruits, meat products, fish products, sweet products, and gold products.

Another important aspect within the scope of the project is the evaluation of marketing campaign responses. The project analyses customer responses to marketing campaigns to identify acceptance patterns and overall campaign effectiveness. This helps in understanding which customer segments are more responsive to marketing efforts and where improvements may be required.

The project also includes analysis of purchase channels such as store purchases, web purchases, and catalog purchases. By comparing different purchase channels, the dashboard helps

organizations identify customer preferences and evaluate the performance of each channel. This insight can support strategic decisions related to marketing and sales channel optimization.

From a technical perspective, the scope of work includes data loading, data transformation, and data modelling using Power BI and Power Query. The cleaned dataset is structured into multiple logical tables, and relationships are established to enable accurate analysis and cross-filtering. Calculated fields and aggregations are created using Power BI's built-in features to support KPI calculations and visualizations.

The project scope also includes the design and development of an interactive dashboard. The dashboard provides key performance indicators, charts, and filters that allow users to explore the data dynamically. Filters are applied across all pages to enable consistent analysis based on selected parameters.

However, certain aspects are intentionally kept outside the scope of this project. The project does not include advanced statistical analysis, machine learning models, or forecasting techniques. It also does not involve real-time data integration or automated data refresh from live systems. The analysis is limited to the provided cleaned dataset and focuses on historical data insights.

3.2 Project Modules

The Marketing Campaign Analysis project is divided into multiple logical modules to ensure systematic development, clarity, and effective implementation. Each module performs a specific function and collectively contributes to achieving the overall objectives of the project. The modular approach improves maintainability, understanding, and scalability of the system.

1. Data Collection Module

This module is responsible for providing the input data required for analysis. The project uses a cleaned marketing campaign dataset that contains customer demographic details, income information, spending behaviour, purchase channel data, and campaign response details. The dataset serves as the foundation of the entire analysis and ensures that accurate and consistent data is available for further processing.

2. Data Preparation and Transformation Module

In this module, the dataset is loaded into Power BI using Power Query. Power Query is used to transform the data by removing unnecessary columns, correcting data formats, and organizing data into a structured form. Calculated columns such as total spending are created during this stage to support meaningful analysis. This module ensures data consistency, accuracy, and readiness for visualization.

3. Data Modelling Module

The Data Modelling module focuses on structuring the data into multiple logical tables to improve analysis efficiency. The cleaned data is divided into separate tables such as Customer, Spending, Campaign, and Purchase Channel tables. Relationships are created between these tables using a unique customer identifier. Proper relationship direction and cardinality are defined to enable accurate filtering and aggregation across the dashboard.

4. Analysis Module

This module performs analytical operations on the structured data to derive insights. It includes analysis of customer demographics, spending patterns across different product categories, campaign response behaviour, and purchase channel preferences. Aggregations and calculations are carried out using Power BI's built-in analytical features to generate meaningful metrics and summaries.

5. Visualization Module

The Visualization module is responsible for presenting insights in a clear and interactive format. Various charts, graphs, tables, and KPI cards are designed using Power BI visuals. These visuals help users understand trends, patterns, and comparisons easily. The dashboard layout is designed to ensure clarity, readability, and effective communication of insights.

6. Interactive Filtering Module

This module enables dynamic data exploration by applying filters across all report pages. Users can filter data based on attributes such as country, education level, marital status, and income range. Cross-filtering between visuals allows users to drill down into specific customer segments and analyse their behaviour in detail.

7. Reporting and Insight Generation Module

The final module focuses on summarizing analytical results and generating business insights. It helps in interpreting dashboard results and identifying actionable outcomes related to marketing campaign effectiveness, customer segmentation, and spending behaviour. These insights can be used by organizations to support strategic marketing decisions.

3.3 Context Diagram (High Level)

A Context Diagram represents the highest-level view of the system and shows how the system interacts with external entities. It provides a simple and clear overview of the system boundary without showing internal processing details. In this project, the context diagram illustrates the interaction between the Marketing Campaign Analysis System and its external stakeholders.

The central system in the context diagram is the Marketing Campaign Analysis Dashboard, developed using Power BI. This system is responsible for processing cleaned marketing data and generating visual insights to support decision-making. The diagram highlights how data flows between the system and external entities.

The primary external entity interacting with the system is the Data Source, which provides the cleaned marketing campaign dataset. This dataset includes customer demographic information, spending data, campaign responses, and purchase channel details. The data source supplies structured input data to the Power BI system for analysis and visualization.

Another important external entity is the Marketing Manager / Business User. This entity represents users who interact with the dashboard to view reports and insights. The marketing manager uses the dashboard to analyse customer behaviour, evaluate campaign performance, and understand spending patterns. The system provides output in the form of interactive dashboards, charts, KPIs, and analytical reports.

The context diagram shows that the system does not directly modify or update the data source. Instead, it consumes the cleaned data as input and performs analysis internally. The output generated by the system is purely analytical and visual, supporting strategic decision-making rather than transactional processing.

By presenting the system at a high level, the context diagram helps stakeholders understand the scope and functionality of the Marketing Campaign Analysis project. It clearly defines system boundaries and demonstrates how the system interacts with external entities without exposing internal complexities.

4 Implementation Methodology

The Implementation Methodology describes the systematic approach followed to design, develop, and implement the Marketing Campaign Analysis project using Power BI. The methodology ensures that the project is executed in an organized manner, starting from data preparation to dashboard development and insight generation.

The implementation of this project follows a structured, phase-wise approach, which helps in maintaining clarity, accuracy, and efficiency throughout the development process.

4.1 Data Acquisition

The first step in the implementation process involves acquiring the marketing campaign dataset. The project uses a cleaned dataset containing customer demographic details, income information, spending patterns across different product categories, purchase channel data, and marketing campaign response information. The dataset serves as the primary input for analysis.

4.2 Data Loading in Power BI

Once the dataset is finalized, it is imported into Power BI using Power Query. Power Query provides an effective interface for loading data from external sources and allows initial inspection of data structure and quality. At this stage, the dataset is reviewed to ensure that all required fields are available for analysis.

4.3 Data Transformation and Preparation

In this phase, Power Query is used to transform the data into a suitable format for analysis. Unnecessary columns are removed, column names are standardized, and data types are verified. Logical tables such as Customer, Spending, Campaign, and Purchase Channel are created to improve data organization. Calculated columns such as total spending are also created during this stage.

4.4 Data Modelling

After data preparation, relationships are established between the logical tables using a unique customer identifier. Proper relationship cardinality and filter direction are defined to ensure accurate data aggregation and cross-filtering. This structured data model forms the backbone of the analytical dashboard.

4.5 Dashboard Design and Visualization

In this stage, various Power BI visuals are designed to present insights effectively. Key Performance Indicators (KPIs) such as total customers, total spending, and average income are displayed using cards. Charts and graphs are used to analyse customer demographics, spending patterns, campaign responses, and purchase channels. The dashboard layout is designed to be user-friendly and visually appealing.

4.6 Interactive Analysis and Filtering

Interactive features such as filters applied across all pages are implemented to allow dynamic data exploration. These filters enable users to analyse specific customer segments based on

parameters such as country, education level, marital status, and income range. Cross-filtering between visuals enhances analytical depth.

4.7 Validation and Testing

Basic validation and testing are performed to ensure accuracy and consistency of the dashboard. This includes verifying data totals, checking filter behaviour, and ensuring that visuals update correctly based on user selections. The testing phase ensures that the dashboard provides reliable insights.

4.8 Insight Generation and Reporting

The final phase involves interpreting the dashboard results and generating meaningful business insights. The analysis focuses on customer behaviour, spending patterns, and campaign effectiveness. These insights can support marketing strategy improvements and data-driven decision-making.

5 Technologies to be used

The successful implementation of the Marketing Campaign Analysis project relies on the use of appropriate software, hardware, and supporting tools. This section highlights the technologies utilized in this project, which ensure efficient data processing, analysis, and visualization.

5.1 Software Platform

The software platform provides a suitable environment for performing data preparation, modelling, and visualization tasks required for this project. The selection of software tools is based on their reliability, ease of use, and suitability for business intelligence and marketing analytics. The primary software tools used in this project are described below:

Power BI Desktop:

Power BI Desktop is a business intelligence tool developed by Microsoft and serves as the core platform for this project. It is used to load, transform, analyse, and visualize the cleaned marketing campaign data. Power BI enables the creation of interactive dashboards and reports through its rich set of visualization features. In this project, Power Query is extensively used for data transformation, while various charts, tables, and KPI cards are used to present insights related to customer behaviour, spending patterns, and campaign performance. The interactive nature of Power BI allows users to explore data dynamically and supports effective decision-making.

Microsoft Excel / CSV:

The marketing dataset used in this project is available in CSV format and is initially reviewed using Microsoft Excel. Excel is used for preliminary data inspection to understand the structure of the dataset and verify data quality. It helps in identifying inconsistencies, missing values, and formatting issues before importing the data into Power BI. This initial review ensures that the dataset is properly cleaned and structured, enabling smooth integration and accurate analysis within Power BI.

5.2 Hardware Platform

The hardware platform provides the physical resources required to run the software tools efficiently. The specifications used in this project include:

- **Processor:** Intel Core i7

- **RAM:** 8 GB minimum
- **Storage:** 500 GB HDD or SSD
- **Operating System:** Windows 11
- **Display:** Minimum resolution 1366 × 768

These specifications ensure smooth data processing, visualization rendering, and interactive dashboard operation without performance lag.

5.3 Tools, if any

In addition to the primary software platforms, several supporting tools and utilities are used in this project to enhance productivity, data quality, and analytical accuracy. These tools play an important role in preparing the data and generating meaningful insights from the marketing campaign dataset.

Power Query:

Power Query, which is integrated within Power BI, is used for data cleaning, transformation, and organization. It enables the removal of unnecessary or redundant columns, renaming of fields for better clarity, correction of data types, and preparation of data in a structured format. Power Query ensures that the cleaned dataset is consistent and ready for accurate analysis within Power BI.

Visualizations and Charts in Power BI:

Power BI provides a wide range of visualization tools that are used extensively in this project. Various chart types such as column charts, bar charts, stacked charts, tables, KPI cards, and summary visuals are used to represent data in a clear and visually appealing manner. These visualizations help in understanding customer behaviour, analysing spending patterns, and evaluating campaign responses effectively.

Data Modelling Tools in Power BI:

The project utilizes Power BI's data modelling capabilities to organize data into multiple logical tables such as Customer, Spending, Campaign, and Purchase Channel. Relationships are defined between these tables using appropriate keys to enable accurate aggregation and cross-filtering across visuals. Proper data modelling improves performance, ensures consistency of results, and enhances the overall analytical capability of the dashboard.

6 Advantages of this Project

The Marketing Campaign Analysis project offers several advantages that enhance the decision-making capabilities of marketing departments and organizations. By transforming raw and cleaned marketing data into an interactive Power BI dashboard, the project delivers both operational and strategic benefits. The key advantages are as follows:

6.1 Data-Driven Decision Making

The project enables organizations to make informed decisions based on actual customer data rather than assumptions. By analysing demographics, spending patterns, and campaign responses, marketing managers can identify target customer segments, allocate resources

efficiently, and design personalized campaigns. This reduces the risk of ineffective marketing efforts and enhances overall business performance.

6.2 Interactive and User-Friendly Dashboard

The use of Power BI to develop an interactive dashboard allows users to explore the data dynamically. Filters such as country, education level, marital status, and income range enable users to focus on specific customer segments. The interactive nature of the dashboard enhances usability, reduces dependency on IT personnel, and allows real-time exploration of marketing insights.

6.3 Efficient Analysis of Customer Behaviour

By segmenting customers based on demographics and purchasing behaviour, the project provides valuable insights into customer preferences. Organizations can identify which groups contribute the most to revenue, which products are preferred, and how different campaigns perform across customer segments. This understanding leads to more effective marketing strategies.

6.4 Simplified Visualization of Key Metrics

The project consolidates multiple metrics into a visually appealing and easily understandable format. Key Performance Indicators (KPIs) such as total customers, total spending, average income, and campaign response rate are displayed prominently. Visual charts and graphs allow stakeholders to quickly grasp trends and patterns without going through complex data tables.

6.5 Time and Cost Efficiency

Manual analysis of large datasets is time-consuming and prone to errors. By automating the analysis and visualization process using Power BI, the project saves significant time and reduces human errors. This allows marketing teams to focus on strategic planning rather than data preparation and calculation.

6.6 Strategic Marketing Insights

The dashboard provides actionable insights such as high-spending customer segments, popular product categories, and campaign effectiveness. These insights help organizations to optimize marketing strategies, improve customer targeting, and enhance return on investment (ROI) for campaigns.

6.7 Flexibility and Scalability

The project is designed to accommodate updates in the dataset. As new marketing data becomes available, it can be integrated into the existing dashboard without major changes. This flexibility ensures that the system remains useful over time and can scale with the organization's data growth.

6.8 Enhanced Reporting and Communication

The interactive dashboard provides a centralized platform for reporting, enabling marketing managers to communicate insights effectively to other stakeholders. The visualization of complex data in simple charts facilitates better understanding and decision-making across teams.

7 Assumptions, if any

The following assumptions have been considered during the implementation of the Marketing Campaign Analysis project. These assumptions define the boundaries and constraints of the analysis and ensure clarity in interpreting the results:

7.1 Cleaned Data is Accurate

It is assumed that the marketing dataset provided has been thoroughly cleaned and validated prior to import into Power BI. All inconsistencies, missing values, and redundant columns have been removed. The project assumes that the data reflects accurate customer information and spending patterns.

7.2 Campaign Responses are Binary

The marketing campaign response data is assumed to be binary, where '1' indicates acceptance of a campaign and '0' indicates non-acceptance. It is assumed that these responses are correctly recorded and represent true customer behaviour.

7.3 No Real-Time Data Integration

The project assumes that the data used is static and historical. There is no integration with real-time data sources. Analysis and insights are generated based on the cleaned dataset provided at the time of implementation.

7.4 Customer Demographics are Fixed

It is assumed that demographic information such as age, education, marital status, income, and country is static for the period under analysis. Any changes in these attributes over time are not reflected in the current dataset.

7.5 Purchase Channel Data is Complete

It is assumed that all information regarding customer purchase channels—web, store, and catalog—is complete and accurately captured. The analysis assumes that these columns correctly reflect the customer's purchasing behaviour.

7.6 Exclusion of Time-Based Analysis

Since the Date/Customer Join column was removed due to errors, it is assumed that the project does not require any time-based or trend analysis. All insights are based on the existing attributes without considering temporal variations.

7.7 Platform and Software Compatibility

It is assumed that the project is implemented using Microsoft Power BI Desktop and that the software and hardware requirements are met. No compatibility issues are present that could affect dashboard functionality.

8 Future Scope and further enhancement of the Project

The Marketing Campaign Analysis project demonstrates the effective use of Power BI to analyse cleaned marketing data and generate meaningful insights. While the current project focuses on descriptive analytics and interactive visualization, there are several opportunities for future scope and enhancements that can further improve the system's capabilities and provide deeper business value.

8.1 Real-Time Data Integration

One potential enhancement is the integration of real-time marketing data into the dashboard. By connecting Power BI to live data sources, such as CRM systems, sales databases, or web analytics platforms, the dashboard could provide up-to-date insights. Real-time data integration would allow marketing managers to track campaign responses, customer behaviour, and spending patterns instantly, facilitating timely and informed decision-making.

8.2 Predictive Analytics and Machine Learning

Currently, the project focuses on descriptive and diagnostic analysis. Future improvements could include predictive analytics using machine learning techniques. For example, predictive models could forecast customer spending behaviour, predict campaign acceptance rates, or identify potential high-value customers. Integrating predictive insights into the dashboard would enable organizations to proactively design targeted marketing strategies.

8.3 Time-Series and Trend Analysis

Although the current implementation does not include temporal analysis, future versions could incorporate time-series data to study trends over months, quarters, or years. Analysing customer spending and campaign responses over time would provide valuable insights into seasonal trends, campaign effectiveness over periods, and the impact of marketing initiatives on different customer segments.

8.4 Expanded Data Sources and Additional Attributes

The current dataset is limited to basic customer demographics, spending behaviour, purchase channels, and campaign responses. In the future, additional attributes such as customer engagement scores, social media interactions, loyalty program data, or geographic details like city and region could be incorporated. Expanded datasets would enable more granular analysis and richer insights.

8.5 Advanced Visualization and Customization

Future enhancements could include more advanced visualizations such as heat maps, tree maps, and interactive drill-through reports. Customizable dashboards for different user roles, such as marketing executives, managers, and analysts, would improve usability and relevance of insights.

8.6 Automated Reporting and Alerts

Another potential enhancement is the creation of automated reporting and alert systems. For instance, Power BI can be configured to send automated reports or notifications to marketing managers when specific KPIs fall below or exceed certain thresholds. This would enhance proactive monitoring and rapid response to marketing performance indicators.

8.7 Integration with Business Intelligence Tools

The project can be further enhanced by integrating with other BI tools or enterprise platforms for a more comprehensive analysis ecosystem. Combining Power BI dashboards with ERP systems, CRM platforms, or cloud-based analytics tools would provide a 360-degree view of customer behaviour and marketing performance.

8.8 Scalability and Cloud Deployment

Future improvements could include deploying the dashboard on cloud platforms for accessibility across multiple locations. Scalability options would allow the system to handle larger datasets efficiently and support collaboration among multiple stakeholders.

9 Project Repository Location

https://github.com/vinayjmd77/Marketing_Campaign_Performance_Analysis_Dashboard

S#	Project Artifacts (softcopy)	Location
1.	Project Synopsis Report (Final Version)	https://github.com/vinayjmd77/Marketing_Campaign_Analysis_Dashboard
2.	Project Progress updates	https://github.com/vinayjmd77/Marketing_Campaign_Analysis_Dashboard
3.	Project Requirement specifications	https://github.com/vinayjmd77/Marketing_Campaign_Analysis_Dashboard
4.	Project Report (Final Version)	https://github.com/vinayjmd77/Marketing_Campaign_Analysis_Dashboard
5.	Test Repository	https://github.com/vinayjmd77/Marketing_Campaign_Analysis_Dashboard
6.	Any other document, give details	https://github.com/vinayjmd77/Marketing_Campaign_Analysis_Dashboard

10 Definitions, Acronyms, and Abbreviations

This section provides definitions of key terms, acronyms, and abbreviations used throughout the Marketing Campaign Analysis project. Understanding these terms ensures clarity and consistency while interpreting the report and dashboard.

Abbreviation	Description
Marketing Campaign	A structured set of promotional activities designed to reach specific customer segments and achieve marketing objectives.
Customer Segmentation	The process of dividing customers into groups based on demographic, behavioural, or spending characteristics for targeted marketing.
Spending Pattern	The analysis of how customers allocate their purchases across different product categories.
Purchase Channel	The medium through which a customer makes a purchase, such as web, store, or catalog.
Dashboard	A visual interface that displays key metrics, charts, and data visualizations to facilitate decision-making.
Interactive Visualization	A feature that allows users to explore and filter data dynamically within a dashboard.
Key Performance Indicator (KPI)	A measurable value that indicates the performance of a business process or marketing campaign.
Data Model	The structured representation of data tables and relationships used to support analysis and reporting.

11 Conclusion

The Marketing Campaign Analysis project successfully demonstrates the use of Power BI for transforming cleaned marketing data into an interactive and insightful dashboard. The project has effectively addressed the challenges associated with analysing large volumes of customer-related data and provided a structured approach for understanding customer behaviour, spending patterns, and marketing campaign responses.

Through the development of logical tables—Customer, Spending, Campaign, and Purchase Channel—and establishing proper relationships between them, the project ensures accurate data aggregation and cross-filtering. Key Performance Indicators (KPIs) such as total customers, total spending, average income, and campaign response rate have been clearly presented, allowing users to quickly assess overall marketing performance.

The dashboard offers interactive features, including filters for country, education level, marital status, and income range, enabling focused analysis on specific customer segments. Charts and visualizations provide a comprehensive view of spending trends, campaign effectiveness, and purchase channel preferences, which help marketing managers make informed, data-driven decisions.

The project also highlights the advantages of using business intelligence tools for marketing analysis. By automating data aggregation and visualization, the dashboard reduces manual effort, minimizes errors, and improves the efficiency of reporting. The insights generated through this project provide a valuable foundation for optimizing marketing strategies, enhancing campaign effectiveness, and improving overall business performance.

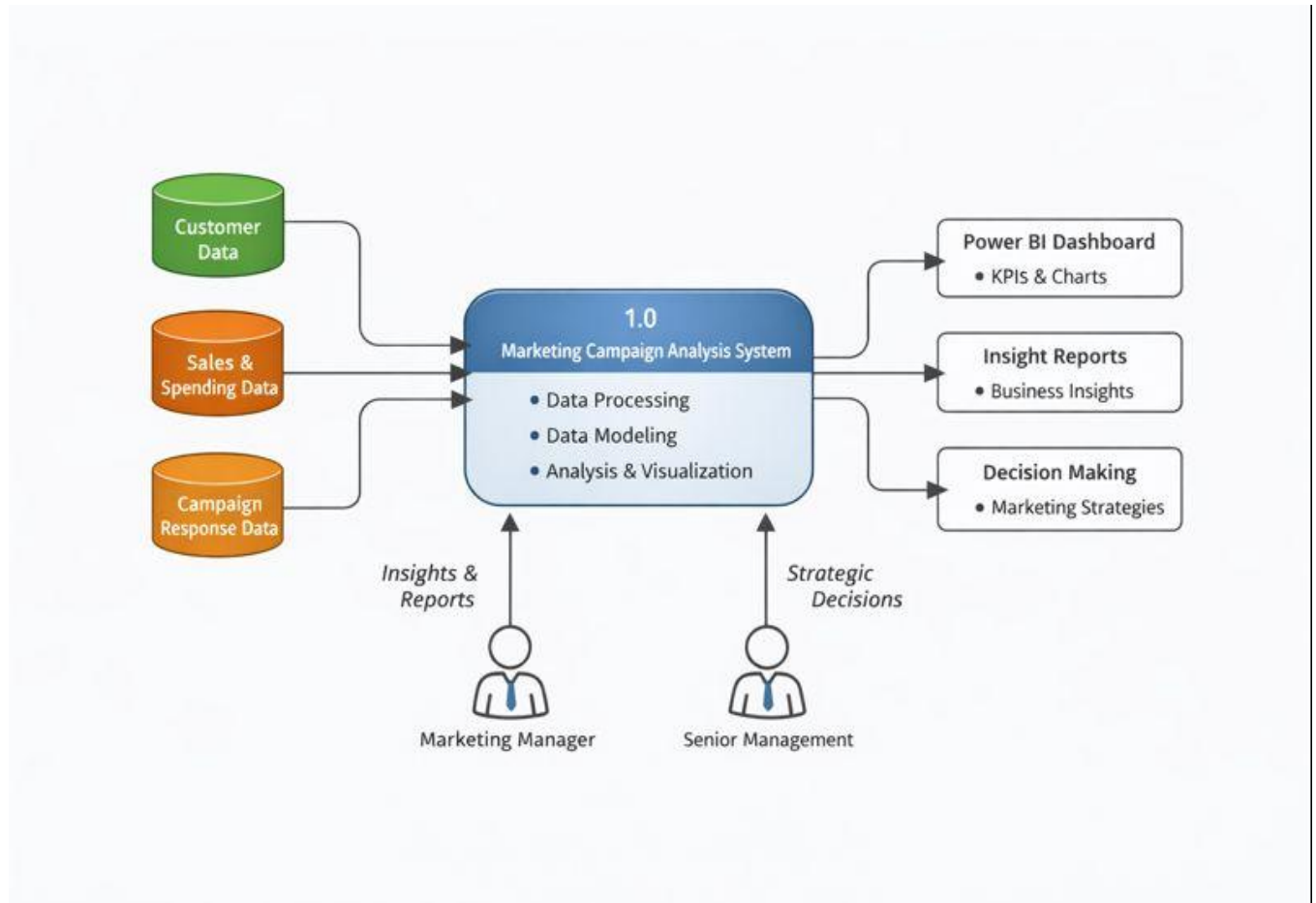
Although certain limitations, such as the absence of time-based trend analysis and predictive modelling, were present due to dataset constraints, the project sets a strong foundation for further enhancements. The future scope includes real-time data integration, predictive analytics, advanced visualizations, and cloud deployment, which can further improve the decision-making capabilities of organizations.

12 References

- Microsoft Power BI Documentation, <https://docs.microsoft.com/en-us/power-bi/>
- Marketing Campaign Dataset
- Power BI Desktop – Data Modeling and Visualization Tutorials, Microsoft Learn, <https://learn.microsoft.com/en-us/power-bi/guided-learning/>

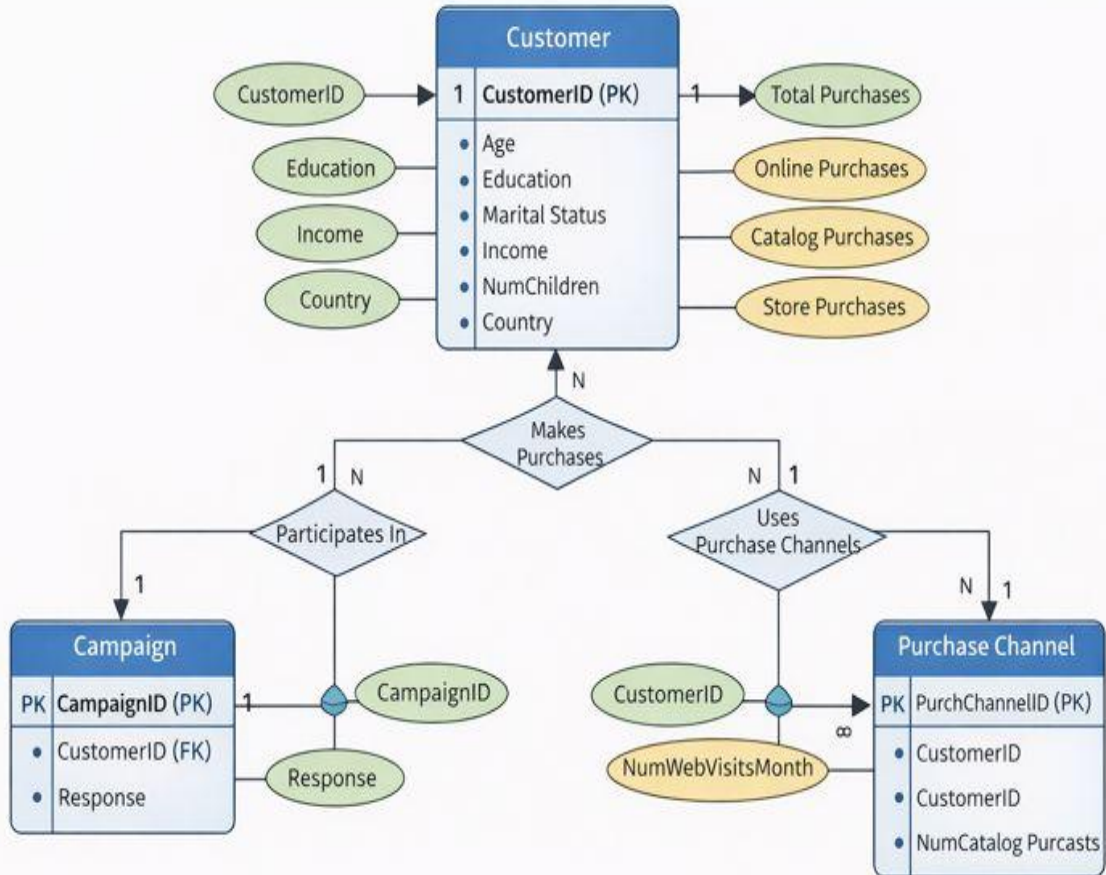
Annexure A

Data Flow Diagram (DFD)



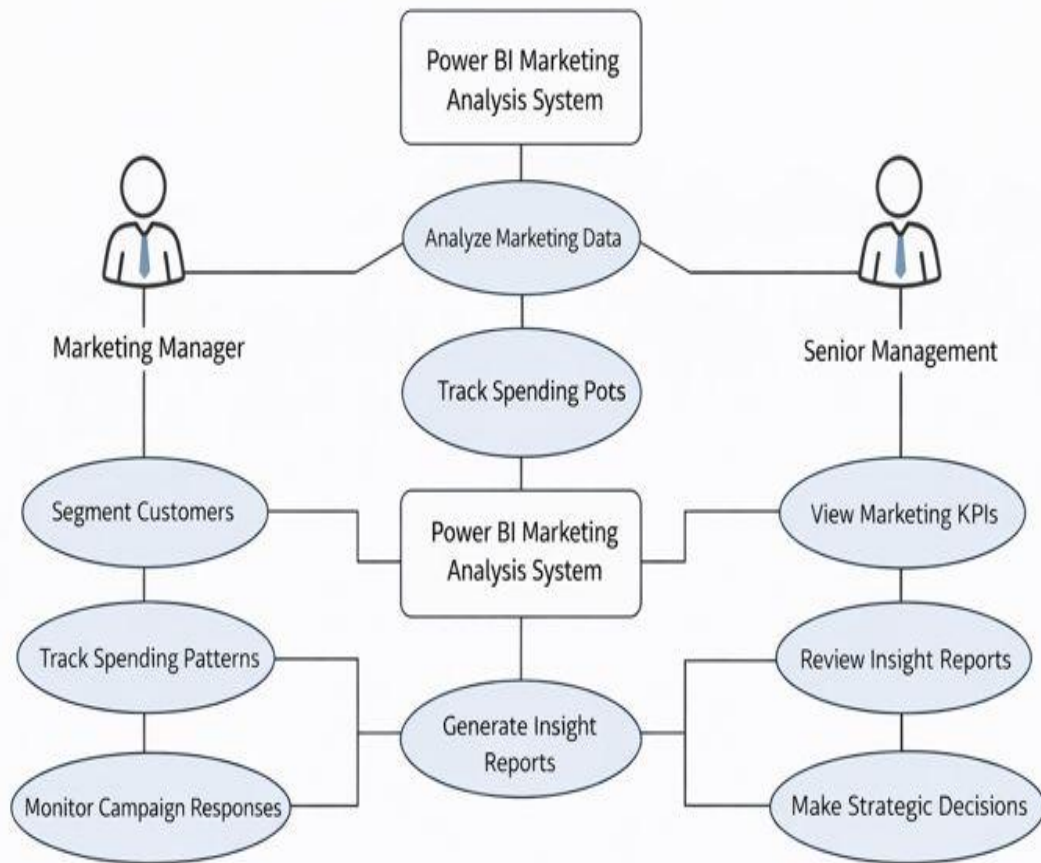
Annexure B

Entity-Relationship Diagram (ERD)



Annexure C

Use-Case Diagram (UCD)



Annexure D

Data Dictionary (DD)

Example:

Customer Table

Fields	Data type	Description
Customer_ID	Number	Unique Customer identification Number
Age	Number	Age Of the Customer
Gender	Text	Gender Of the Customer
Marital_status	Text	Marital Status
Education	Text	Customer education level
Income	Number	Annual income of the customer
Kidhome	Number	Number of kids in house land
Teenhome	Number	Number of teenagers in house land
Recency	Number	Days since last purchase
Enrollment_date	Date	Date when customer joined

Purchase table:

Fields	Data type	Description
Customer_id	Number	Customer reference id
NumDealsPurchase	Number	Purchase made during deals
NumWebPurchases	Number	Purchase made through websites
NumCatalogPurchases	Number	Purchase made with catalogs
NumStorePurchase	Number	Purchase with store
NumWebVisitMonth	Number	Websites visit in last month

Spending Table

Fields	Data type	Description
Customer_id	Number	Customer reference id
MntWines	Number	Money spends on wines
MntFruits	Number	Money spends on fruits
MntMeatProducts	Number	Money spends on meat products
MntFishProducts	Number	Money spends on fish products
MntSweettProducts	Number	Money spends on sweets
MntGoldProducts	Number	Money spends on gold products
TotalSpending	Number	Total amount spends across all categories

Campaign Response Table

Fields	Data type	Description
Customer_id	Number	Customer reference id
AcceptedCmp1	Boolean	Accepted campaign 1
AcceptedCmp2	Boolean	Accepted campaign 2
AcceptedCmp3	Boolean	Accepted campaign 3
AcceptedCmp4	Boolean	Accepted campaign 4
AcceptedCmp5	Boolean	Accepted campaign 5
Response	Boolean	Weather customer accepted the last campaign

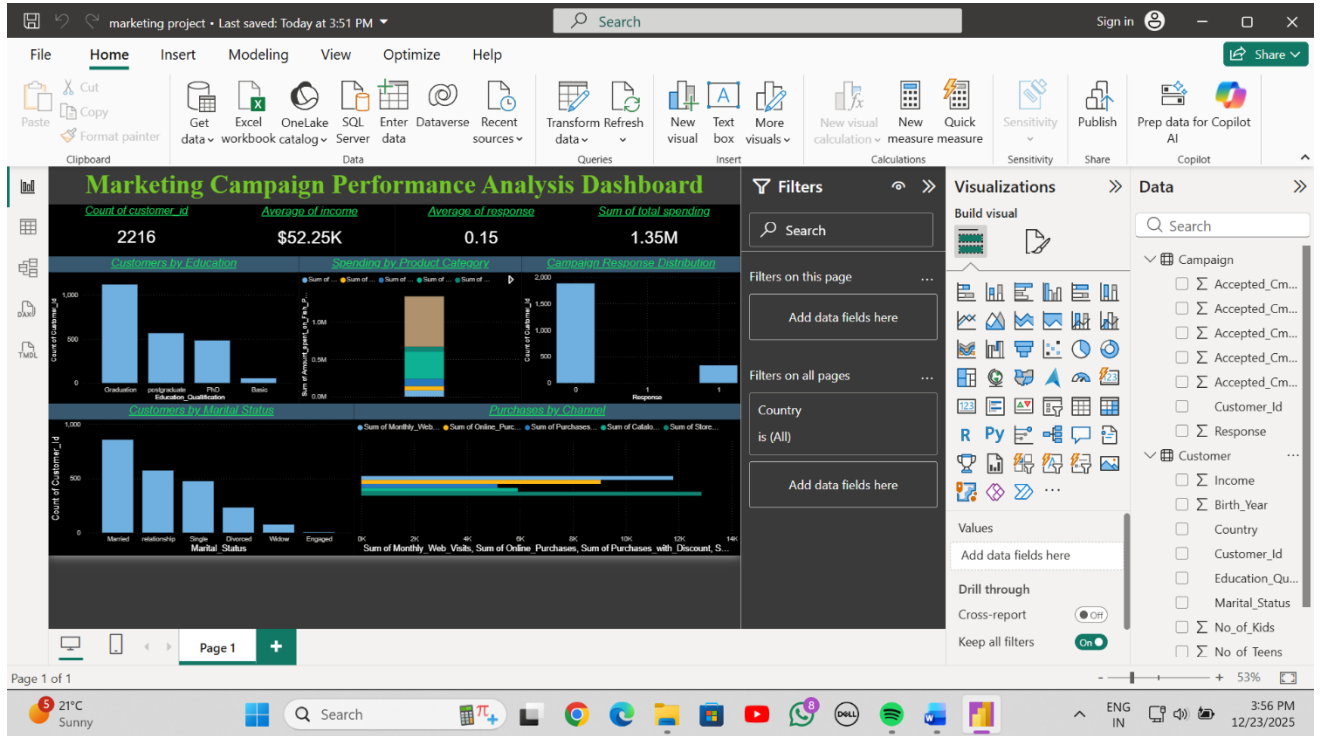
Geography Table

Fields	Data type	Description
Customer_id	Number	Customer reference id
Country	Text	Customer country
City	Text	Customer city
Region	Text	Geographical region

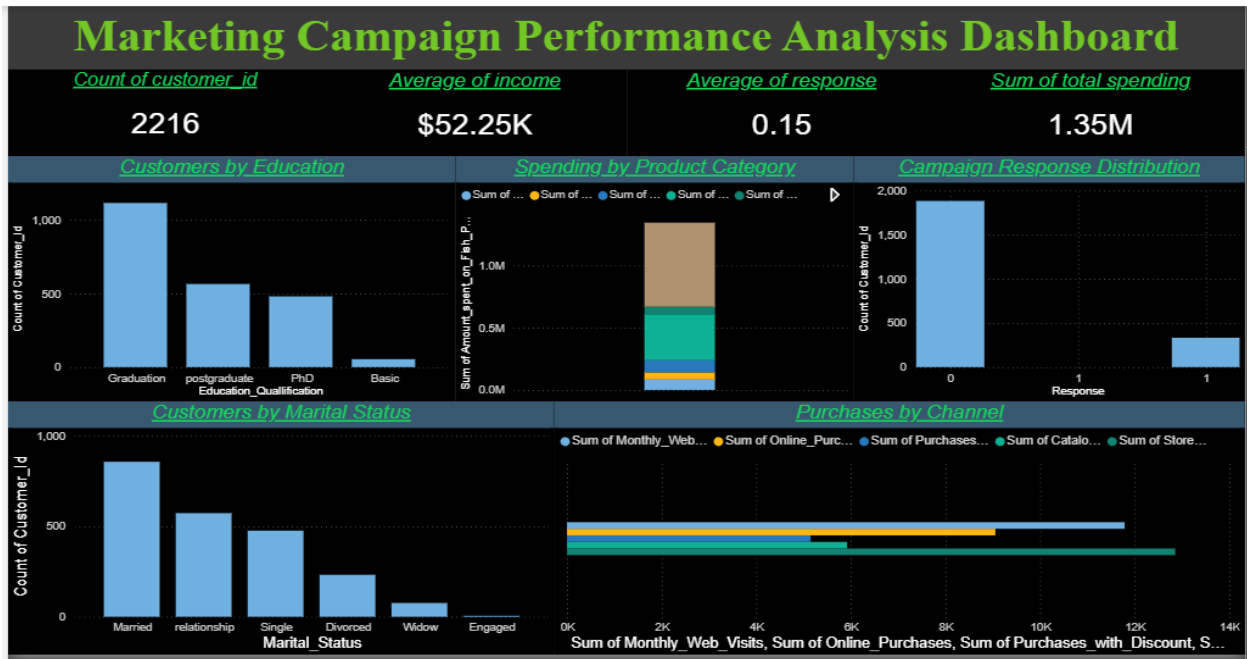
Annexure E

Screen Shots

Home Page:



Dashboard:



Model View:

